MATH 128 - HW8 - Jie Wen

Problem 1: Problems from the textbook, chapter 3.1

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b) $f(x) = \sqrt{1+x}$

$f(0) = 1, f(0.6) = \sqrt{1.6} \approx 1.264911064, f(0.9) = \sqrt{1.9} \approx 1.378404875.$

Degree ≤1 (through x_0, x_1)

$$P_1(x) = 1 + \frac{\sqrt{1.6} - 1}{0.6} x \approx 1 + 0.4415184401x.$$

$$P_1(0.45) \approx 1.198683298.$$

Degree ≤2 (through x_0, x_1, x_2)

$$P_2(x) \approx -0.070228595896x^2 + 0.483655597650x + 1.$$

$$P_2(0.45) \approx 1.203423728274.$$

True value and absolute errors (at $x = 0.45$): $f(0.45) = \sqrt{1.45} \approx 1.204159457879.$

$$|f - P_1| \approx 5.476159828716 \times 10^{-3}, \qquad |f - P_2| \approx 7.357296057142 \times 10^{-4}.$$

c) $f(x) = \ln(1+x)$

$f(0) = 0, f(0.6) = \ln(1.6) = 0.470003629246, f(0.9) = \ln(1.9) = 0.641853886172.$

Degree ≤1 (through x_0, x_1)

$$P_1(x) = 0 + \frac{0.470003629246 - 0}{0.6} x \approx 0.783339382076x.$$

$$P_1(0.45) \approx 0.352502721934.$$

Degree ≤2 (through x_0, x_1, x_2)

$$P_2(x) \approx -0.233894658134x^2 + 0.923676176957x + 0.$$

$$P_2(0.45) \approx 0.368290611358.$$

True value and absolute errors (at $x = 0.45$): $f(0.45) = \ln(1.45) \approx 0.371563556432.$

$$|f - P_1| \approx 1.906083449818 \times 10^{-2}, \qquad |f - P_2| \approx 3.272945074129 \times 10^{-3}.$$

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Using the Lagrange remainder,

$$|f(x) - P_n(x)| = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i) \right| \quad \text{for some } \xi \in [x_0, x_n],$$

so a practical bound is

$$|f(x) - P_n(x)| \le \frac{\max\limits_{t \in [x_0, x_n]} |f^{(n+1)}(t)|}{(n+1)!} \left| \prod_{i=0}^{n} (x - x_i) \right|.$$

Here $x_0 = 0, x_1 = 0.6, x_2 = 0.9$ and the evaluation is at $x = 0.45$.

Geometry terms at $x = 0.45$

- Linear ($n = 1$): $(x - x_0)(x - x_1) = (0.45)(-0.15) = -0.0675 \Rightarrow |\cdot| = 0.0675$, divide by 2! $\Rightarrow 0.0675/2 = 0.03375$.
- Quadratic ($n = 2$): $(x - x_0)(x - x_1)(x - x_2) = (0.45)(-0.15)(-0.45) = 0.030375 \Rightarrow |\cdot| = 0.030375$, divide by 3! $\Rightarrow 0.030375/6 = 0.0050625$.

b) $f(x) = \sqrt{1+x}$

$$f''(x) = \frac{-1}{4(1+x)^{3/2}}, \qquad f^{(3)}(x) = \frac{3}{8(1+x)^{5/2}}.$$

Over $[0, 0.9]$,

$$\max|f''| = \frac{1}{4} \quad (\text{at } x=0), \qquad \max|f^{(3)}| = \frac{3}{8} \quad (\text{at } x=0).$$

Bounds at $x = 0.45$:

- Linear ($n=1$):

$$|f - P_1| \le \frac{\max|f''|}{2}|(x-x_0)(x-x_1)| = \frac{1/4}{2} \cdot 0.0675 = 0.0084375.$$

- Quadratic ($n=2$):

$$|f - P_2| \le \frac{\max|f^{(3)}|}{6}|(x-x_0)(x-x_1)(x-x_2)| = \frac{3/8}{6} \cdot 0.030375 = 0.0018984375.$$

(These bounds dominate the actual errors from the previous step, which were ~5.48×10^{-3} and 7.36×10^{-4}.)

c) $f(x) = \ln(1+x)$

$$f''(x) = -\frac{1}{(1+x)^2}, \qquad f^{(3)}(x) = \frac{2}{(1+x)^3}.$$

Thus on $[0, 0.9]$:
$\max|f''| = 1$ and $\max|f^{(3)}| = 2$ (both at $x=0$).

Bounds at $x = 0.45$:

- Linear ($n=1$):

$$|f - P_1| \le \frac{1}{2} \cdot 0.0675 = 0.03375.$$

- Quadratic ($n=2$):

$$|f - P_2| \le \frac{2}{6} \cdot 0.030375 = 0.010125.$$

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Degree 1 (linear) using $(-0.25, 0.25)$

At $x=0$: $L_0(0) = L_1(0) = \frac{1}{2}$.

$$P_1(0) = \tfrac{1}{2}f(-0.25) + \tfrac{1}{2}f(0.25) = \frac{1.33203 + 0.800781}{2} = 1.066406 \text{ (approx)}$$

Degree 2 (quadratic) using $(-0.5, -0.25, 0.25)$

Basis weights at $x=0$: $\{-\frac{1}{3}, 1, \frac{1}{3}\}$.

$$\begin{aligned} P_2(0) &= -\tfrac{1}{3}f(-0.5) + 1 \cdot f(-0.25) + \tfrac{1}{3}f(0.25) \\ &= -\tfrac{1}{3}(1.93750) + 1.33203 + \tfrac{1}{3}(0.800781) = 0.953124 \text{ (approx)} \end{aligned}$$

Degree 3 (cubic) using all four nodes $(-0.5, -0.25, 0.25, 0.5)$

Basis weights at $x=0$: $\{-\frac{1}{6}, \frac{2}{3}, \frac{2}{3}, -\frac{1}{6}\}$.

$$\begin{aligned} P_3(0) &= -\tfrac{1}{6}f(-0.5) + \tfrac{2}{3}f(-0.25) + \tfrac{2}{3}f(0.25) - \tfrac{1}{6}f(0.5) \\ &= -\tfrac{1}{6}(1.93750) + \tfrac{2}{3}(1.33203) + \tfrac{2}{3}(0.800781) - \tfrac{1}{6}(0.687500) \\ &= 0.984374 \text{ (approx)} \end{aligned}$$

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Because $f(0) = f(1) = 0$, the quadratic Lagrange interpolant through $x_0 = 0, x_1, x_2 = 1$ reduces to the single middle basis term:

$$P_2(x) = f(x_1)L_1(x), \qquad L_1(x) = \frac{(x-0)(x-1)}{(x_1-0)(x_1-1)}.$$

Evaluate at $x = \frac{1}{2}$:

$$P_2\left(\tfrac{1}{2}\right) = f(x_1)\frac{\frac{1}{2}(\frac{1}{2}-1)}{x_1(x_1-1)} = f(x_1)\frac{\frac{1}{4}}{x_1(1-x_1)}.$$

Since $f(x) = \sqrt{x(1-x)}$, write $t := x_1(1-x_1) \in (0, \frac{1}{4}]$. Then

$$f\left(\tfrac{1}{2}\right) - P_2\left(\tfrac{1}{2}\right) = \tfrac{1}{2} - \frac{\frac{1}{4}\sqrt{t}}{t} = \tfrac{1}{2} - \frac{1}{4\sqrt{t}}.$$

Set this equal to $-\frac{1}{4}$ and solve:

$$\tfrac{1}{2} - \frac{1}{4\sqrt{t}} = -\tfrac{1}{4} \Longrightarrow t = \frac{1}{9}.$$

So $x_1(1-x_1) = \frac{1}{9}$, i.e.

$$x_1^2 - x_1 + \frac{1}{9} = 0 \Longrightarrow x_1 = \frac{1}{2} \pm \frac{\sqrt{5}}{6}.$$

The largest solution in $(0,1)$ is

$$x_1 = \frac{1}{2} + \frac{\sqrt{5}}{6}.$$

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b) $f(x) = \sin(\ln x)$, nodes $x_0 = 2.0, x_1 = 2.4, x_2 = 2.6$ ($n = 2$)

Interpolant $P_2(x)$ (Lagrange form)

Let

$$\begin{aligned} L_0(x) &= \frac{(x-2.4)(x-2.6)}{(2.0-2.4)(2.0-2.6)} = \frac{(x-2.4)(x-2.6)}{0.24}, \\ L_1(x) &= \frac{(x-2.0)(x-2.6)}{(2.4-2.0)(2.4-2.6)} = \frac{(x-2.0)(x-2.6)}{-0.08}, \\ L_2(x) &= \frac{(x-2.0)(x-2.4)}{(2.6-2.0)(2.6-2.4)} = \frac{(x-2.0)(x-2.4)}{0.12}. \end{aligned}$$

Then

$$P_2(x) = \sin(\ln 2.0)L_0(x) + \sin(\ln 2.4)L_1(x) + \sin(\ln 2.6)L_2(x).$$

Power-basis:

$$P_2(x) \approx -0.13063441x^2 + 0.89699789x - 0.63249687.$$

Error bound on $[2.0, 2.6]$

$n = 2$, so now need $|f^{(3)}(x)|$ on the interval. For $f(x) = \sin(\ln x)$,

$$f'(x) = \frac{\cos(\ln x)}{x}, \quad f''(x) = -\frac{\sin(\ln x) + \cos(\ln x)}{x^2}, \quad f^{(3)}(x) = \frac{3\sin(\ln x) + \cos(\ln x)}{x^3}.$$

Hence $|f^{(3)}(x)| \le \frac{\sqrt{3^2+1^2}}{x^3} = \frac{\sqrt{10}}{x^3}$, so on $[2, 2.6]$,
$\max |f^{(3)}| \le \sqrt{10}/8$.

Let $w(x) = (x-2.0)(x-2.4)(x-2.6)$. Over $[2, 2.6]$, $\max|w(x)| \approx 0.0169009$ (attained near $x \approx 2.15696$). Therefore

$$|R_2(x)| \le \frac{\max|f^{(3)}|}{3!} \max|w(x)| \le \frac{\sqrt{10}/8}{6}(0.0169009) \approx 1.11 \times 10^{-3}.$$

c) $f(x) = \ln x$, nodes $x_0 = 1, x_1 = 1.1, x_2 = 1.3, x_3 = 1.4$ ($n = 3$)

Interpolant $P_3(x)$ (Lagrange form)

$$P_3(x) = \sum_{k=0}^{3} \ln(x_k) L_k(x), \quad L_k(x) = \prod_{\substack{i=0 \\ i \neq k}}^{3} \frac{x - x_i}{x_k - x_i}.$$

With the given nodes:

$$\begin{aligned} x_0 = 1: &\quad (1-1.1)(1-1.3)(1-1.4) = -0.012, \\ x_1 = 1.1: &\quad (1.1-1)(1.1-1.3)(1.1-1.4) = +0.006, \\ x_2 = 1.3: &\quad (1.3-1)(1.3-1.1)(1.3-1.4) = -0.006, \\ x_3 = 1.4: &\quad (1.4-1)(1.4-1.1)(1.4-1.3) = +0.012, \end{aligned}$$

so explicitly

$$P_3(x) = \ln(1)\frac{(x-1.1)(x-1.3)(x-1.4)}{-0.012} + \ln(1.1)\frac{(x-1)(x-1.3)(x-1.4)}{0.006} + \ln(1.3)\frac{(x-1)(x-1.1)(x-1.4)}{-0.006} + \ln(1.4)\frac{(x-1)(x-1.1)(x-1.3)}{0.012}.$$

Power-basis:

$$P_3(x) \approx 0.19700561x^3 - 1.06259032x^2 + 2.53245290x - 1.66686819.$$

Error bound on $[1, 1.4]$

Here $n = 3$, so

$$R_3(x) = \frac{f^{(4)}(\xi)}{4!}(x-1)(x-1.1)(x-1.3)(x-1.4).$$

For $f(x) = \ln x$, $f^{(4)}(x) = -\dfrac{6}{x^4}$; hence $\max_{[1,1.4]} |f^{(4)}| \le 6$.
Let $w(x) = \prod_{i=0}^{3}(x - x_i)$. On $[1, 1.4]$, $\max |w(x)|$ occurs at $x = 1.2$ with $|w(1.2)| = 0.0004$. Thus

$$|R_3(x)| \le \frac{6}{4!}0.0004 = 1.0 \times 10^{-4}.$$

Problem 2: Problems from the textbook, chapter 3.2

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Neville's method applied at the target x for each dataset: use the standard recursion $Q_{i,j}(x) = \dfrac{(x - x_{i-j})Q_{i,j-1}(x) - (x - x_i)Q_{i-1,j-1}(x)}{x_i - x_{i-j}}$, built column-by-column from $Q_{i,0} = f(x_i)$.

a) $f(x) = 3^x$, nodes $(-2, -1, 0, 1, 2)$; approximate $\sqrt{3} = 3^{1/2}$ at $x = 0.5$

$f(-2) = \frac{1}{9}, f(-1) = \frac{1}{3}, f(0) = 1, f(1) = 3, f(2) = 9$.

Running Neville's table up to $Q_{4,4}(0.5)$ gives

$$\sqrt{3} \approx P(0.5) = 1.7083333333333335,$$

Actual $\sqrt{3} \approx 1.7320508075688772$, so

$$\text{abs. error} = |1.7083333333333335 - 1.7320508075688772| = 2.3717474236 \times 10^{-2}.$$

b) $f(x) = \sqrt{x}$, nodes $(0, 1, 2, 4, 5)$; approximate $\sqrt{3}$ at $x = 3$

Values: $f(0) = 0, f(1) = 1, f(2) = \sqrt{2}, f(4) = 2, f(5) = \sqrt{5}$.

Neville up to $Q_{4,4}(3)$ gives

$$\sqrt{3} \approx P(3) = 1.6906067646231164,$$

Actual $\sqrt{3} \approx 1.7320508075688772$, so

$$\text{abs. error} = |1.6906067646231164 - 1.7320508075688772| = 4.1444042946 \times 10^{-2}.$$

c) Which is more accurate?

(2.37×10^{-2}) vs (4.14×10^{-2}), **part (a) is better**.

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$x = 0.4, x_0 = 0, x_1 = 0.25, x_2 = 0.5, x_3 = 0.75$.
Neville's recurrence (for entries at the same x) is

$$P_{i,i+1}(x) = \frac{(x - x_i)P_{i+1} - (x - x_{i+1})P_i}{x_{i+1} - x_i}, \qquad P_{i,i+1,i+2}(x) = \frac{(x - x_i)P_{i+1,i+2} - (x - x_{i+2})P_{i,i+1}}{x_{i+2} - x_i}, \text{ etc.}$$

1. From the 2-point step with $i = 1$:

$$P_{1,2} = \frac{(0.4 - 0.25)P_2 - (0.4 - 0.5)P_1}{0.5 - 0.25} = \frac{0.15P_2 + 0.2}{0.25} = 0.6P_2 + 0.8.$$

2. From the 3-point step using the given $P_{2,3} = 2.4$ and $P_{1,2,3} = 2.96$:

$$P_{1,2,3} = \frac{(0.4 - 0.25)P_{2,3} - (0.4 - 0.75)P_{1,2}}{0.75 - 0.25} = \frac{0.36 + 0.35P_{1,2}}{0.5} = 0.72 + 0.7P_{1,2}.$$

   Thus $2.96 = 0.72 + 0.7P_{1,2} \Rightarrow P_{1,2} = 3.2$.
3. Equate with step (1): $3.2 = 0.6P_2 + 0.8 \Rightarrow 0.6P_2 = 2.4 \Rightarrow P_2 = 4$.